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**PROJECT TITLE : PHARMACY AND PRESCRIPTION
MANAGEMENT SYSTEM (PPMS)**

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EXECUTIVE SUMMARY

The Pharmacy and Prescription Management System (PPMS) is a database-oriented application designed to streamline and automate the critical operations of a pharmacy. The central objective is to transition from manual record-keeping and siloed systems, which currently lead to data duplication, inaccuracies, and inefficiency, to a cohesive, modernized platform. This system will emphasize the secure and effective management of medications, prescriptions, patient details, and sales data. A key technical specification is the PPMS's ability to demonstrate database interoperability and portability by being compatible with and utilizing PostgreSQL, MySQL, and Microsoft SQL Server. This cross-platform capability not only enhances operational flexibility but also serves as a valuable case study in effective database design and execution across various platforms.

The project's scope is defined by the development of five highly integrated modules designed to solve specific operational pain points. The Medicine Inventory Management Module will accurately track stock levels, expiry dates, and supplier information to eliminate stock shortages and errors. The Prescription Management Module is crucial for allowing pharmacists to efficiently create, edit, and manage prescriptions, thereby reducing the miscommunication and delays currently plaguing doctor-pharmacist interactions. Furthermore, the Patient Information Management Module will securely store and update comprehensive patient profiles and medical records, directly resolving issues of incomplete histories faced with manual storage. The system will also feature a Sales and Billing Module to automate transaction recording and invoice generation, making the checkout process faster and less error-prone.

Finally, the PPMS will integrate a powerful Reports and Analytics Module that generates comprehensive data on stock, sales trends, prescriptions, and overall system usage. This feature is vital for better decision-making and for the pharmacy to monitor performance, which is currently a challenge due to the lack of proper tools. The system is intended for use by Pharmacists, Patients, and System Administrators, making it a comprehensive solution for digital healthcare management. By implementing this system using HTML, CSS, JavaScript, and Node.js, the team aims to deliver a modern application that not only improves the operational effectiveness of pharmacies but also enhances healthcare information management strategies by setting a standard for accuracy and platform flexibility.

1 CHAPTER 1: INTRODUCTION

1.1 Introduction

The Pharmacy and Prescription Management System (PPMS) is a database-oriented application created to streamline and automate the functions of a pharmacy. It emphasizes the secure and effective management of medications, prescriptions, patient details, and sales data. In conventional pharmacies, many records are still managed manually or through unconnected systems, which often result in data inconsistencies, delays, and mistakes. This initiative seeks to develop a cohesive system that can function with various database systems, specifically PostgreSQL, MySQL, and SQL Server. This requirement is intended to showcase database interoperability and portability. The system will assist pharmacists, doctors, and patients by delivering precise, immediate access to information for managing prescriptions, monitoring inventory, and processing billing. The PPMS project aims to provide a modern solution that enhances healthcare information management and operational effectiveness.

1.2 Problem Statements

The Pharmacy and Prescription Management System (PPMS) project is motivated by several critical issues present in conventional pharmacy operations that rely on manual record-keeping or unconnected systems. The primary problems this project seeks to resolve are:

- Inventory Errors and Stock Management Issues.

Manual processes in recording and tracking medicine stock, expiry dates, and supplier details often cause inventory errors and stock shortages.

- Disintegrated Prescription Handling.

The absence of an integrated prescription management system leads to confusion, delayed updates, and miscommunication between doctors and pharmacists.

- Incomplete Patient Records.

Storing patient information manually results in incomplete records, making it difficult to retrieve accurate medical histories.

- Slow and Error-Prone Sales Process.

The sales and billing process is slow and prone to human error due to the lack of an automated transaction recording system.

- Lack of Performance Monitoring.

Without proper reporting and analytics tools, the pharmacy struggles to monitor sales trends, medicine performance, and overall system efficiency.

1.3 Objectives

The objectives of the Pharmacy and Prescription Management System (PPMS) project are directly aligned with resolving the identified problems by developing five highly functional and integrated modules. The specific objectives are:

- To develop a Medicine Inventory Management Module that accurately tracks medicine details, quantities, expiry dates, and supplier information.
- To design a Prescription Management Module that allows pharmacists to create, edit, and manage prescriptions efficiently within the system.
- To build a Patient Information Management Module that securely stores and updates patient profiles and medical records.
- To implement a Sales and Billing Module that automates invoice generation and transaction recording for efficient sales management.
- To create a Reports and Analytics Module that generates comprehensive reports on stock, sales, prescriptions, and system usage for better decision-making.

1.4 Scope of the Project

1.4.1. Module to be developed

- i. Medicine Inventory Management Module: Manages medicine details, stock levels, expiry dates, and suppliers.
- ii. Prescription Management Module: Allows pharmacists to create and update prescriptions linked to patients and medicines.
- iii. Patient Information Management Module: Stores and updates patient profiles and medical histories.
- iv. Sales and Billing Module: Generates invoices and maintains sales records.
- v. Reports and Analytics Module: Produces comprehensive reports on sales, stock, prescriptions, and system usage.

1.4.2. Target Users

The system is designed to serve and be utilized by three main groups:

- 1) Pharmacists
- 2) Patients
- 3) System Administrators

1.4.3. Technical Requirements

A critical technical requirement for this project is to demonstrate database interoperability. The system will be built to function with multiple database systems , specifically utilizing and showcasing compatibility across PostgreSQL, MySQL, and Microsoft SQL Server. The application development will be based on technologies including HTML, CSS, JavaScript, and Node.js.

1.5 Project Significances

The Pharmacy and Prescription Management System (PPMS) project holds significant importance on both operational and educational levels.

- Improved Operational Effectiveness.

The initiative is primarily motivated by the necessity for precise and effective data management in pharmacies, particularly in settings where mistakes can directly impact patient well-being. The PPMS will minimize human mistakes, enhance healthcare information management, and improve the operational efficiency of pharmacies.

- Showcasing Database Interoperability.

The project acts as a useful examination of how database systems can be created to function on various platforms, which is important information for database administrators and developers. The project seeks to showcase effective database design and execution on various platforms by demonstrating compatibility with PostgreSQL, MySQL, and SQL Server.

- Practical Learning Experience.

Through the adoption of this system, students will acquire practical experience with various Database Management Systems (DBMSs), database architecture, and actual system development. This aids in the enhancement of healthcare data management strategies.

- Conforming to Standards.

The PPMS is designed to conform to contemporary digital healthcare management standards.

1.6 Conclusion

Chapter 1 successfully introduces the Pharmacy and Prescription Management System (PPMS), defining it as a database-oriented application focused on modernizing pharmacy operations. The chapter clearly establishes the critical problems in current

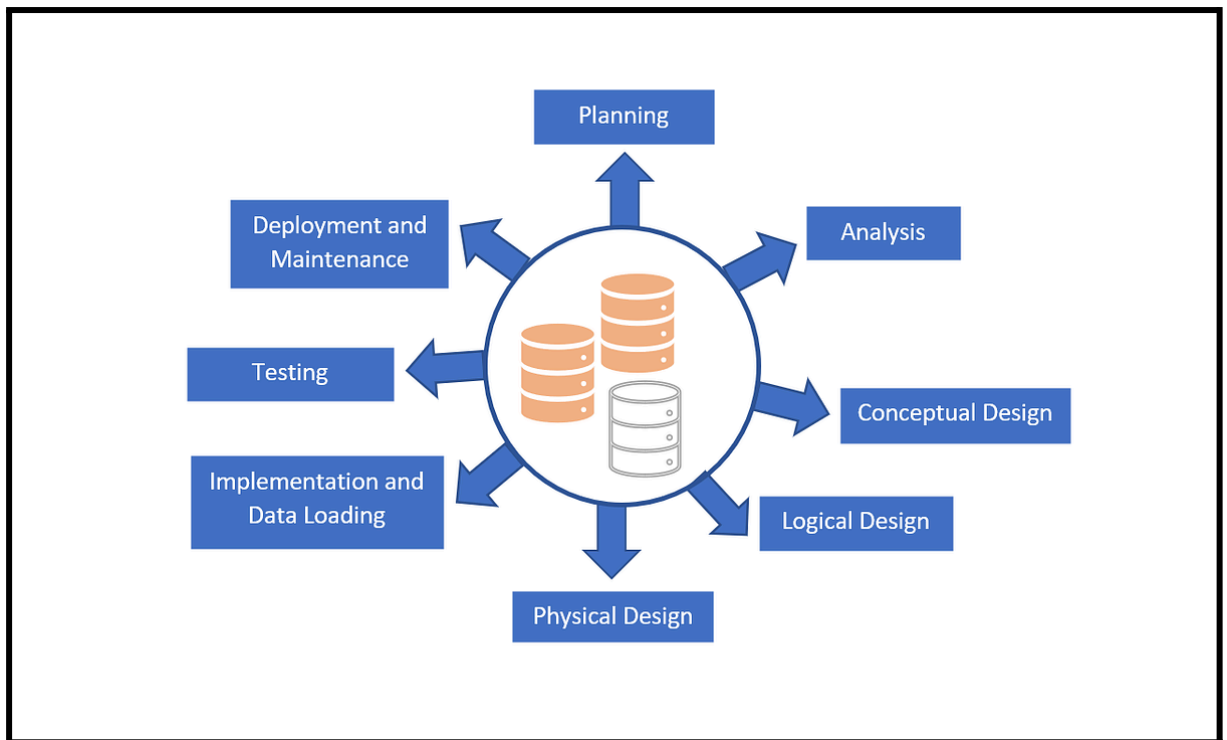
manual or disconnected systems, such as inventory errors and delayed communications. Furthermore, it sets forth five precise, module-based objectives to resolve these issues. Finally, the scope is clearly delineated, emphasizing the development of five core modules, serving three target user groups, and demonstrating essential database interoperability across PostgreSQL, MySQL, and SQL Server. This foundation solidifies the necessity and direction of the PPMS project.

2 CHAPTER 2: DATABASE SYSTEM METHODOLOGY AND PLANNING

2.1 Introduction

This chapter, Database System Methodology and Planning, serves as the blueprint for the systematic execution of the Pharmacy and Prescription Management System (PPMS) project. Given the project's complexity, which includes developing five core functional modules and, critically, ensuring database interoperability across multiple platforms (PostgreSQL, MySQL, and SQL Server), a rigorous methodology and detailed planning are paramount to success. This section will first detail the chosen Database Development Methodology, justifying why a structured approach is necessary for maintaining data integrity and system reliability, while an iterative component ensures timely delivery and testing of the application layer. Subsequently, it will outline the principles of Project Management and Requirement handling, including the breakdown of tasks, resource allocation, and strategy for managing both functional and non-functional requirements. Finally, a clear set of Project Milestones will be defined, establishing a timeline against which progress will be measured to ensure the PPMS is delivered successfully, on time, and strictly within the defined scope. This planning framework aims to mitigate risks, ensure resource optimization, and guide the team through the challenges inherent in multi-platform database application development.

2.2 Database Development Methodology



The Pharmacy and Prescription Management System (PPMS) adopts the Database Life Cycle (DBLC) as the main methodology for database development in this project. The DBLC provides a structured approach to obtaining requirements, analysing data needs, designing the database, implementing it on multiple Database Management Systems (DBMSs), and maintaining it after deployment. The life cycle moves to the next stage only after the previous one has been completed and validated, which helps ensure that the PPMS database is consistent, reliable, and aligned with the system objectives.

The DBLC used in this project consists of six main stages: Database Initial Study, Database Design, Implementation and Loading, Testing and Evaluation, Operation, and Maintenance and Evolution. Each stage is described below in the context of the PPMS.

i. Database Initial Study

In this phase, the mission and objectives of the PPMS database are defined. The scope and boundaries of the project are identified by analysing the current pharmacy environment, including manual processes for inventory tracking, prescription handling, patient record storage, and sales recording. This study helps to identify key entities such as medicines, patients, prescriptions, users, and sales, as well as the data structures and relationships needed to support them. The main problems and constraints, such as inventory errors, incomplete

patient histories, slow billing, and the need for interoperability across PostgreSQL, MySQL, and SQL Server, are also documented in this phase.

ii. Database Design

The second stage focuses on designing the PPMS database model to support secure and efficient pharmacy operations. This phase covers three levels of design:

- Conceptual design – development of the Entity-Relationship Diagram (ERD), describing entities (e.g. MEDICINE, PATIENT, PRESCRIPTION), their attributes, and relationships in a technology-independent form.
- Logical design – transformation of the ERD into normalized tables with clearly defined primary keys, foreign keys, and constraints, forming the data dictionary for PPMS.
- Physical design – definition of the actual storage structures and Data Definition Language (DDL) statements for each target DBMS (PostgreSQL, MySQL, SQL Server), including data types, indexes, and integrity constraints.

During this stage, the choice and configuration of DBMS software are also planned to ensure that the same logical model can be deployed consistently across multiple platforms.

iii. Implementation and Loading

This phase is the physical realization of the PPMS database design. The logical schema and DDL scripts are used to create the database objects—such as tables, views, indexes, stored procedures, and triggers—inside each selected DBMS. After the schemas are created, initial and sample data for medicines, patients, prescriptions, users, and sales are inserted into the tables to support development and testing activities. Application-level components (e.g. forms and modules for inventory, prescriptions, and billing) are then implemented to interact with the database using appropriate query languages and database drivers.

iv. Testing and Evaluation

Testing and evaluation ensure that the PPMS database and related application components function according to the specified requirements. This includes:

- Verifying that all tables, relationships, and constraints are created correctly and enforce data integrity.

- Testing stored procedures, triggers, and views to ensure that business rules—such as stock updates after sales or validation of prescription data—are executed correctly.
- Running queries and transactions from the PPMS interface to confirm that CRUD (Create, Read, Update, Delete) operations behave as expected.
- Evaluating performance and compatibility across PostgreSQL, MySQL, and SQL Server by executing equivalent operations and comparing the results.

Any issues identified during this stage lead to adjustments in the design or implementation before moving to the next stage.

v. Operation

Once the PPMS database has successfully passed testing, it is considered ready for operational use. In this stage, the database, the PPMS application, and its users (pharmacists, administrators, and patients) form a complete information system. Real pharmacy workflows are carried out using the system, including recording prescriptions, updating stock, processing sales, and generating reports. Operational data flows—such as daily sales transactions, prescription records, and inventory movements—are handled directly by the PPMS database in each deployed DBMS environment.

vi. Maintenance and Evolution

The final stage ensures that the PPMS database remains reliable, secure, and up-to-date over time. Regular maintenance tasks include:

- Preventive and corrective maintenance – performing backup and recovery, monitoring performance, and fixing data or schema issues when they arise.
- Adaptive maintenance – modifying the schema, queries, or stored procedures to accommodate new requirements, regulations, or changes in pharmacy workflows.
- User and access management – creating and updating user accounts, roles, and permissions for pharmacists, administrators, and other stakeholders.

The database administrator is responsible for planning and executing these maintenance activities to preserve data integrity and support the continuous evolution of the PPMS as the pharmacy's needs grow.

2.3 Project Management and Requirement

No	Activity / Task	Start Date	End Date	Dur.	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
1	Discussion topics proposals (Students & Supervisors)	06/10/2025	17/10/2025	12																	
2	Preparing Proposal Report	13/10/2025	24/10/2025	12																	
3	Methodology & Database System Planning Reports	20/10/2025	07/11/2025	19																	
4	Develop system and database design	03/11/2025	21/11/2025	19																	
5	Implementation of Database Systems	17/11/2025	19/12/2025	33																	
6	Integration, Testing & Poster Design	08/12/2025	19/12/2025	12																	
7	Draft Final Report, Poster & Logbook	15/12/2025	09/01/2026	26																	
8	Workshop 2 Showcase & Peer Assessment	05/01/2026	16/01/2026	12																	
9	Final Report and Logbook Submission	12/01/2026	23/01/2026	12																	

Project management for the PPMS is based on breaking down the work into clear tasks aligned with the main modules and database activities, supported by simple scheduling tools such as Gantt charts or task lists. Functional requirements, such as inventory tracking and prescription management, and non-functional requirements, such as performance, security, and interoperability, are documented and traced to ensure that each requirement is addressed by corresponding design and implementation tasks. Responsibilities are distributed among team members according to expertise, with regular discussions and reviews to monitor progress, address risks, and incorporate feedback from testing.

2.4 Project Milestone

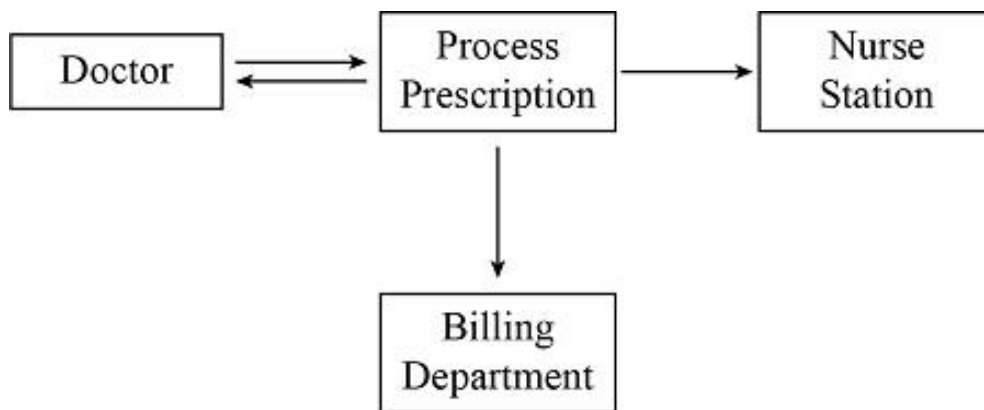
No	Activity / Task	Start Date	End Date	Dur.	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17
1	Discussion topics proposals (Students & Supervisors)	06/10/2025	17/10/2025	12																	
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3 CHAPTER 3: DATABASE SYSTEM AND ANALYSIS

3.1 Introduction

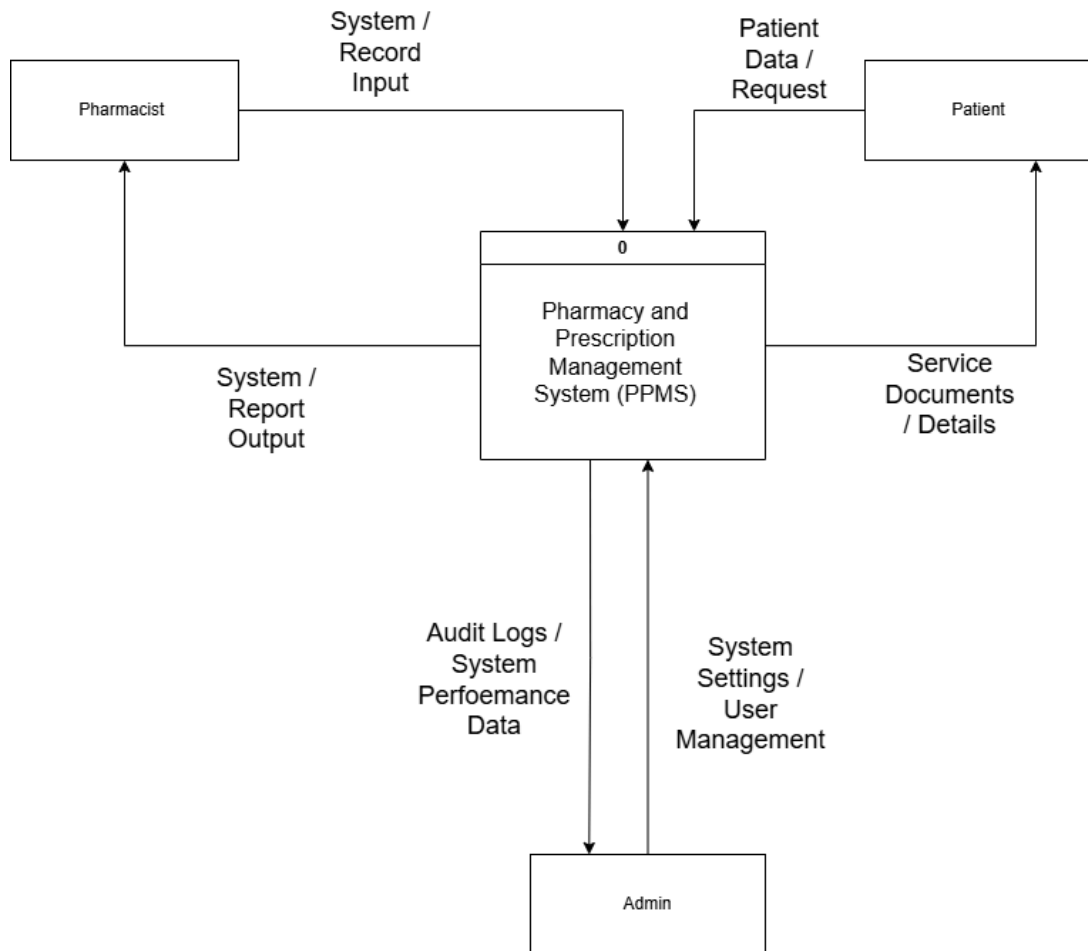
This chapter focuses on the analysis of the existing pharmacy environment and the definition of the “to-be” system realized through the PPMS. It describes the current issues faced by pharmacies, explains the proposed system requirements, and provides visual representations of the system boundaries and data flows through the Context Diagram and Data Flow Diagram (DFD).

3.2 Current System Analysis



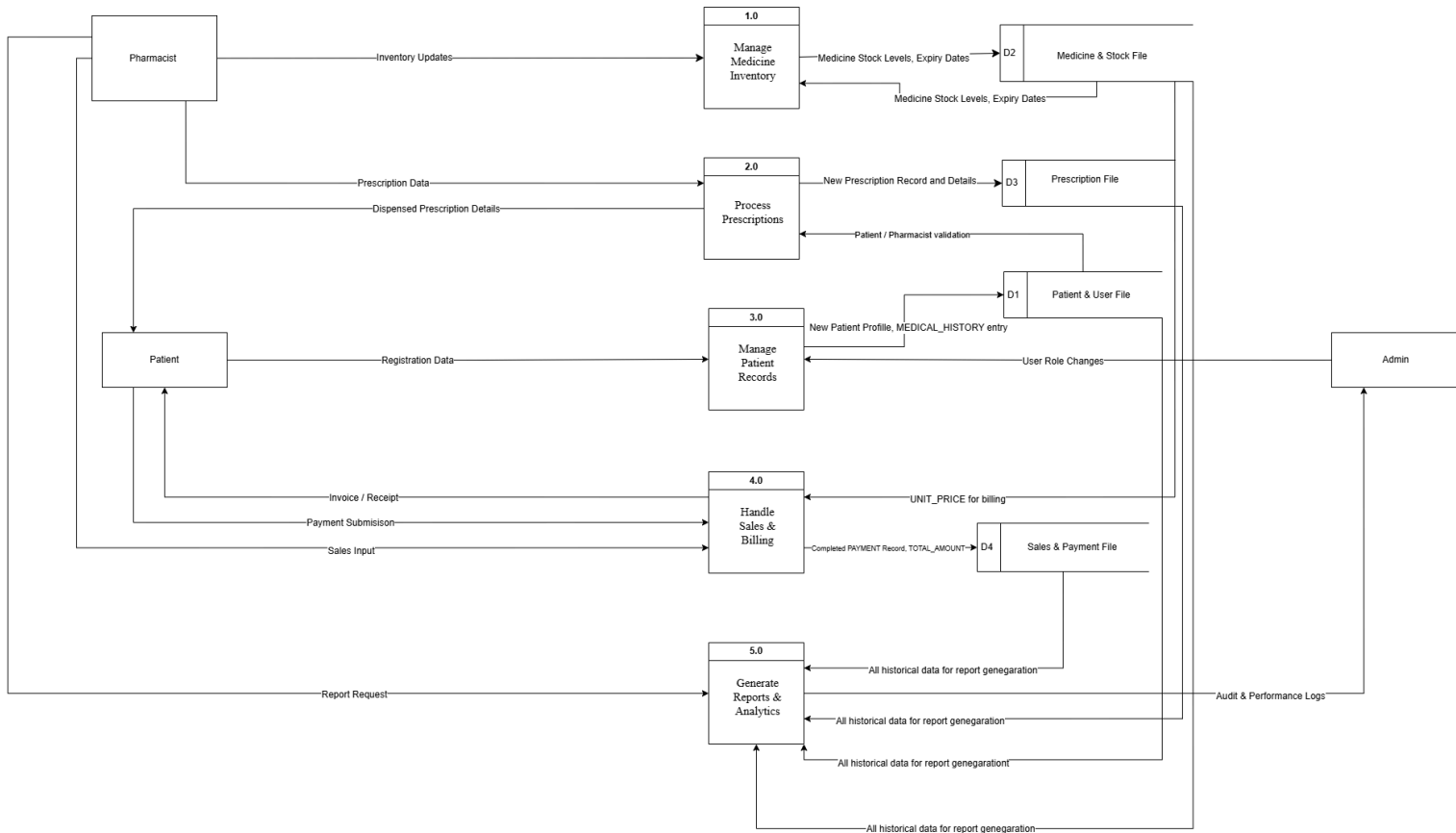
3.3 System To-Be Analysis

3.3.1 Context Diagram



3.3.2 Data Flow Diagram

Data Flow Diagram Level 1



Data Flow Diagram Level 1

3.4 Conclusion

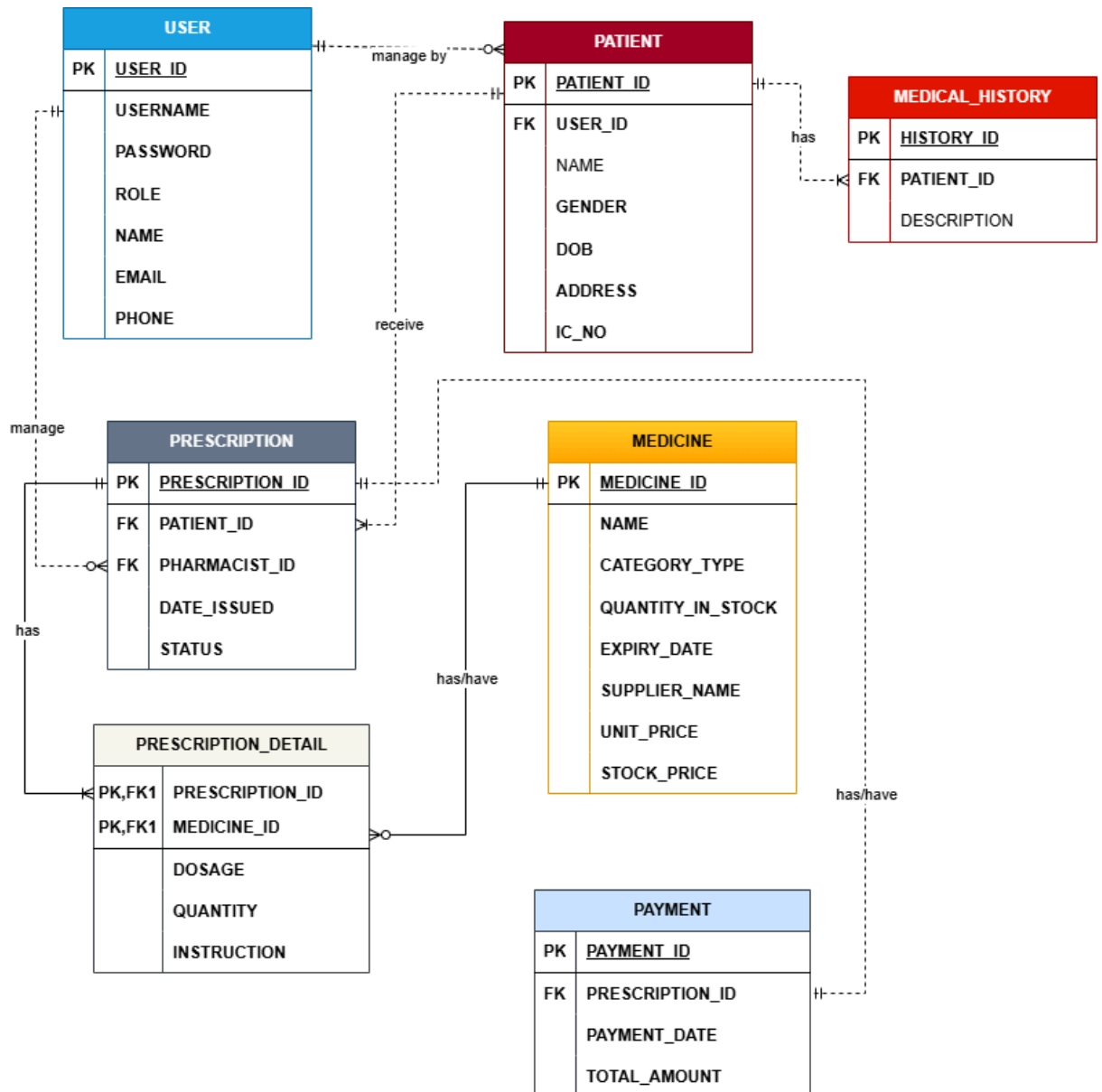
In summary, Chapter 3 clarifies the shortcomings of the current manual or semi-manual systems and defines the requirements of the to-be PPMS through structured analysis. By using the Context Diagram and DFD, the chapter provides a clear view of system boundaries, major processes, data stores, and interactions, which forms a strong basis for the subsequent database design activities.

4 CHAPTER 4: DATABASE DESIGN

4.1 Introduction

This chapter documents the database design for the PPMS, covering conceptual, logical, and physical levels. The goal is to produce a robust, normalized schema that supports the functional requirements of the system while remaining compatible with PostgreSQL, MySQL, and SQL Server.

4.2 Conceptual Database Design



4.3 Logical Database Design

4.3.1 Data Dictionary

4.4 Physical Database Design

4.4.1 Data Definition Language

4.5 Conclusion

5 CHAPTER 5: INDIVIDUAL APPLICATION MODULES AND IMPLEMENTATION

5.1 Introduction

After obtaining the requirement and designing the component to show flow of the system, project implementation is where the project execution is taking place from the documented data into a real program. The focus of this chapter is to show the programming technique used to develop the system and the error handling. Programming technique is the technique used to describe the development process and the details of programming language code used.

5.2 Database Setup and Installation

5.3 Database Implementation

[May include your stored procedures, trigger any related database programming technique including error handling]

5.4 Application Modules

[Module 1, Module 2, Module , Module 4]

[Briefly describe your individual modules]

5.5 User Interface

5.6 Conclusion

6 CHAPTER 6: DATABASE INTEGRATION AND TESTING

6.1 Introduction

This chapter explains how the Pharmacy and Prescription Management System (PPMS) database is installed, integrated, and tested across the three supported Database Management Systems (DBMS): PostgreSQL, MySQL, and Microsoft SQL Server. Database integration in PPMS focuses on deploying the same logical schema on each platform and ensuring that all modules—inventory, prescriptions, patient records, sales, and reports—can access and manipulate data consistently regardless of the underlying DBMS. The chapter also describes the testing activities carried out to verify data integrity, business rules, and performance, including the execution of structured test cases on tables, constraints, stored procedures, and triggers. In addition, database management and administration tasks such as backup, restore, and user access control are documented to demonstrate how the PPMS database can be maintained in an operational environment.

6.2 Database Setup and Installation

[how do you install & configure your database]

6.3 Database Integration and Testing

[how do you perform your database integration & testing]

6.4 Database Management and Administration

[includes how do you perform backup & restore]

6.5 Conclusion

7 CHAPTER 7: CONCLUSION

7.1 Introduction

In conclusion, the Pharmacy and Prescription Management System (PPMS) has been developed to meet the objectives defined at the beginning of this project. Each module and database component was designed to follow proper database development standards while supporting safe and efficient pharmacy operations. Before adoption, the core functions of the system were tested to ensure that prescriptions, inventory, patient information, and sales can be processed correctly. Throughout this report, every aspect of the PPMS that ranging from problem analysis, database design, multi-DBMS implementation, up to integration and testing that has been discussed in detail so that users and stakeholders can clearly understand how the system operates. The challenges faced during development, especially related to interoperability across PostgreSQL, MySQL, and SQL Server, have led to valuable learning and a more robust solution. Ultimately, the PPMS is expected to help pharmacies manage medicines and prescriptions more accurately while saving time and effort for pharmacists and patients.

7.2 Achievement and Application in SDG

At the start of the project, several key modules were identified as necessary to solve the problems found in the existing pharmacy environment. These include the Medicine Inventory Management Module, Prescription Management Module, Patient Information Management Module, Sales and Billing Module, and Reports and Analytics Module. All of these modules have been successfully implemented and integrated with the underlying database architecture. The system is capable of running on three different DBMS platforms like PostgreSQL, MySQL, and SQL Server and demonstrating the intended database interoperability. The objectives and scope defined in Chapter 1 have therefore been fulfilled, as the system can now handle inventory records, prescriptions, patient profiles, and sales transactions in a structured and reliable way. Functional testing carried out in the implementation and testing

chapters shows that the main features work as intended within the defined project constraints.

7.3 Project Limitation

Although the PPMS achieves its core goals, several limitations remain. First, the current version focuses on a single-pharmacy environment and does not yet support multi-branch operations or real-time synchronization between outlets. Second, security features such as encryption of sensitive data, detailed audit trails, and advanced access control are kept at a basic level suitable for an academic project rather than a production healthcare system. Third, the reporting and analytics functions are limited to summary reports and do not yet offer advanced data visualisation or predictive analysis. In addition, while the system is compatible with three DBMS platforms, performance and stress testing under high transaction loads have been minimal, so scalability for large pharmacies has not been fully validated. These limitations provide clear directions for future enhancement of the PPMS.

7.4 Suggestion and Improvement

Several improvements can be implemented in future iterations of the PPMS. The system can be extended to support multi-branch pharmacies with centralized stock control and consolidated reporting. Security can be strengthened by introducing role-based access control, password policies, data encryption, and more comprehensive logging of user activities. The Reports and Analytics Module can be upgraded with interactive dashboards, trend analysis, and forecasting features to assist managers in making data-driven decisions. Integration with external systems, such as e-prescription platforms, barcode scanners, or online payment gateways, would also make the system more practical in real pharmacy settings. Finally, developing a responsive web or mobile interface would allow pharmacists and patients to access key functions more conveniently, further increasing the usability and impact of the PPMS.

7.5 Potential Commercialization

The current implementation of the PPMS is primarily designed as an academic project, but it has clear potential for real-world use, especially in small and medium-sized community pharmacies. With its modular structure and ability to run on widely used DBMS platforms, the system could be packaged as an affordable solution for pharmacies that currently rely on manual records or simple spreadsheets. To be fully commercialised, additional work would be required to harden security, comply with healthcare regulations, improve user experience, and provide comprehensive technical support. If these enhancements are made and the system is tested extensively in real operational environments, the PPMS could evolve into a practical product that supports accurate medicine management, smoother prescription handling, and better decision-making for pharmacy owners.

7.6 Conclusion

Pharmacy and prescription management systems play a critical role in ensuring that medicines are dispensed safely and that pharmacy operations run smoothly. A poorly designed system can lead to inventory errors, delayed treatments, and difficulties in tracking patient records, which may negatively affect both patients and pharmacists. Through this project, the PPMS has addressed key issues identified in the problem statements by providing structured modules for inventory, prescriptions, patient information, sales, and reporting, all supported by a well-designed database that can operate on multiple DBMS platforms. The planned modules have been implemented, the main objectives and scope have been achieved, and the system has undergone functional and integration testing to minimise operational problems. Overall, the project can be considered a success, providing both a working prototype for pharmacy operations and a strong foundation for future enhancement and possible commercial deployment.

